FINAL-YEAR DISSERTATION · BSc (Hons) · NIBM / COVENTRY UNIVERSITY

A Survival-Analysis Framework for Predicting and Explaining Time-to-Employment for Final-Year IT Undergraduates in Sri Lanka

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885
graduates modelled

245
employers

324
real job ads

295
automated tests

The story in six steps

ROADMAP

1 Problem & gap

Why this research?

2 Method & data

How it was built

3 Models & evaluation

How it was tested

4 Results

What was found

5 The artifact

App, ChatGPT, usability

6 Reflection

Limits, contribution, next

The problem: no graduate can see their own timeline

PROBLEM

17,000

IT & computing graduates a year in Sri Lanka (SLASSCOM, 2024)

1 in 4

still searching when surveyed: they are 'not yet', not 'failed'

Rates only

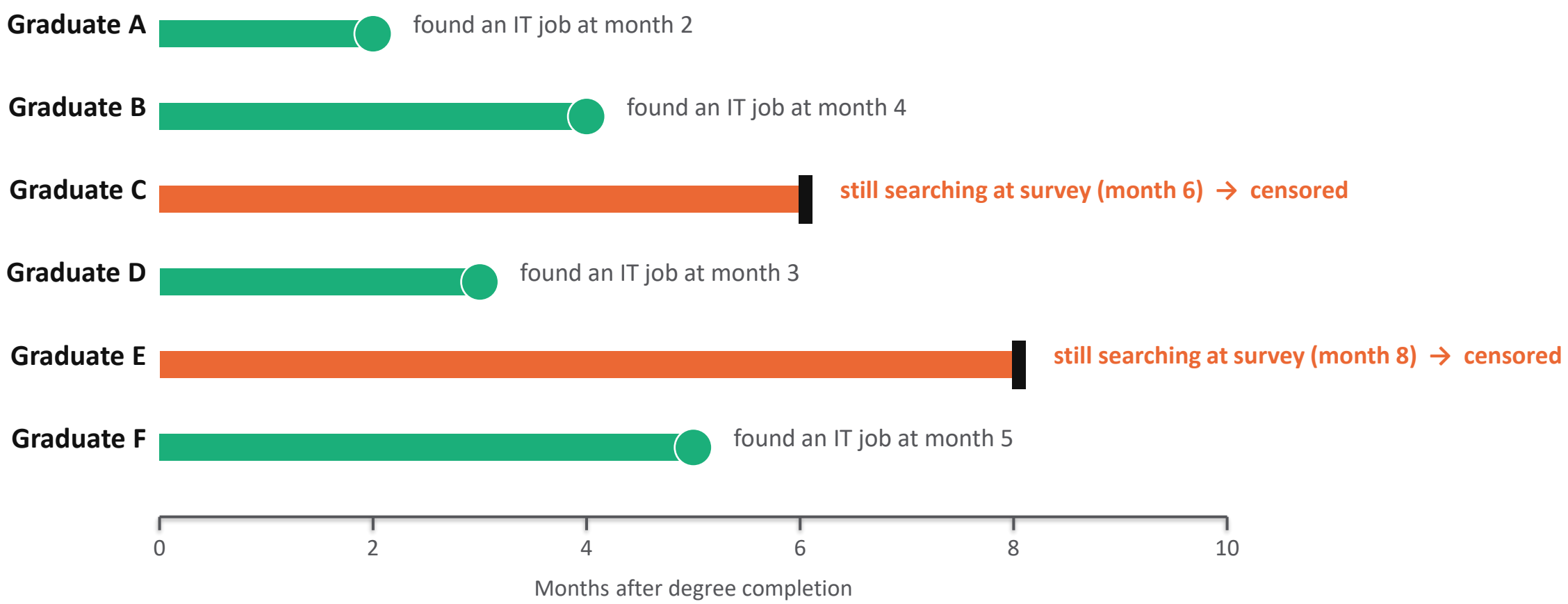
national data report unemployment rates, not individual time to a first job

What a final-year student actually faces today

- **No personal estimate**: 'How long will I take?' is answered with guesses and averages
- **Existing tools**: classify 'employable / not', or match jobs: they ignore WHEN, and discard graduates still searching
- **Generic AI advice**: ChatGPT can talk about careers, but is it right for Sri Lankan graduates? (tested later in this talk)

Why survival analysis? Because 'still searching' is information

PROBLEM



A classifier discards C and E, or mislabels them. Survival analysis uses them, and predicts WHEN, not just whether.

Literature gap: nothing combines these capabilities

PROBLEM

Capability	Rukais et al. 2025	ITConnect	Scopira	Commercial tools	This study
Sri Lankan IT focus	Yes	Yes	No	No	Yes
Predicts time to a job	No	No	No	No	Yes
Uses graduates still searching	No	No	No	No	Yes
Range with uncertainty	No	No	No	No	Yes
Individual explanation + what-if	No	No	Partial	Partial	Yes

Closest local precedent: Rukais et al. (2025), an ensemble model over 820 IT employee profiles that predicts a career-path category, not time.

Gap: an explainable, time-to-event framework for Sri Lankan IT graduates, linked to real employer demand, and tested against a general LLM.

Objectives, research questions, and how each is answered

PROBLEM

	Research question	Answered by
O1 / RQ1	Which academic, technical, professional and experiential factors are associated with time-to-employment?	Kaplan-Meier, log-rank, Cox / AFT time ratios
O2 / RQ2	How well do graduate profiles align with employer requirements and job-market demand?	Graduate vs employer ratings, 324 job ads
O3 / RQ3	How accurately can a survival framework predict time-to-employment, and how do linear, regularised, parametric and nonlinear models compare?	7 models, locked test set, calibration, fairness
O4 / RQ4	Does the prototype give accurate, explainable, actionable guidance (usability and a benchmark against a general LLM)?	SUS with 12 users, ChatGPT benchmark

Research design at a glance

METHOD

1 Collect

graduates, employers, job ads

2 Clean & audit

rules fixed, every removal logged

3 Split once

70 / 15 / 15, stratified

4 Fit 7 models

on training only

5 Select

pre-set rule on validation

6 Test once

locked, never re-tuned

7 Explain

time ratios, what-ifs

8 Deploy

app + API, nothing stored

- **Pragmatist philosophy, deductive and quantitative**; the plan, metric and selection rule were fixed before analysis

Data: three sources, triangulated

METHOD

1,175

graduate responses (47 items): outcome + profile + 26 skill ratings

245

employer responses (21 items): importance of skills and hiring factors

324

real entry-level IT job ads from 211 employers (12 Jul - 11 Oct 2026)

- **Primary + secondary**: questionnaires (Google Forms) plus real job advertisements
- **Sampling**: purposive, convenience and snowball: non-probability, because no sampling frame of IT graduates exists
- **Sample size**: Cochran's formula $n = Z^2p(1-p)/e^2 = 384$ (95%, ±5%, $p = 0.5$) → target about 400; collected 1,175
- **Ethics**: ethical approval before collection; consent on every response; no names or contacts; Sri Lanka PDPA 2022

Instrument quality: validity and reliability

METHOD

α 0.925

technical skills (17 items), 95% CI 0.916-0.933

α 0.902

soft skills (9 items), 95% CI 0.890-0.913

KMO 0.95

sampling adequacy of both skill blocks

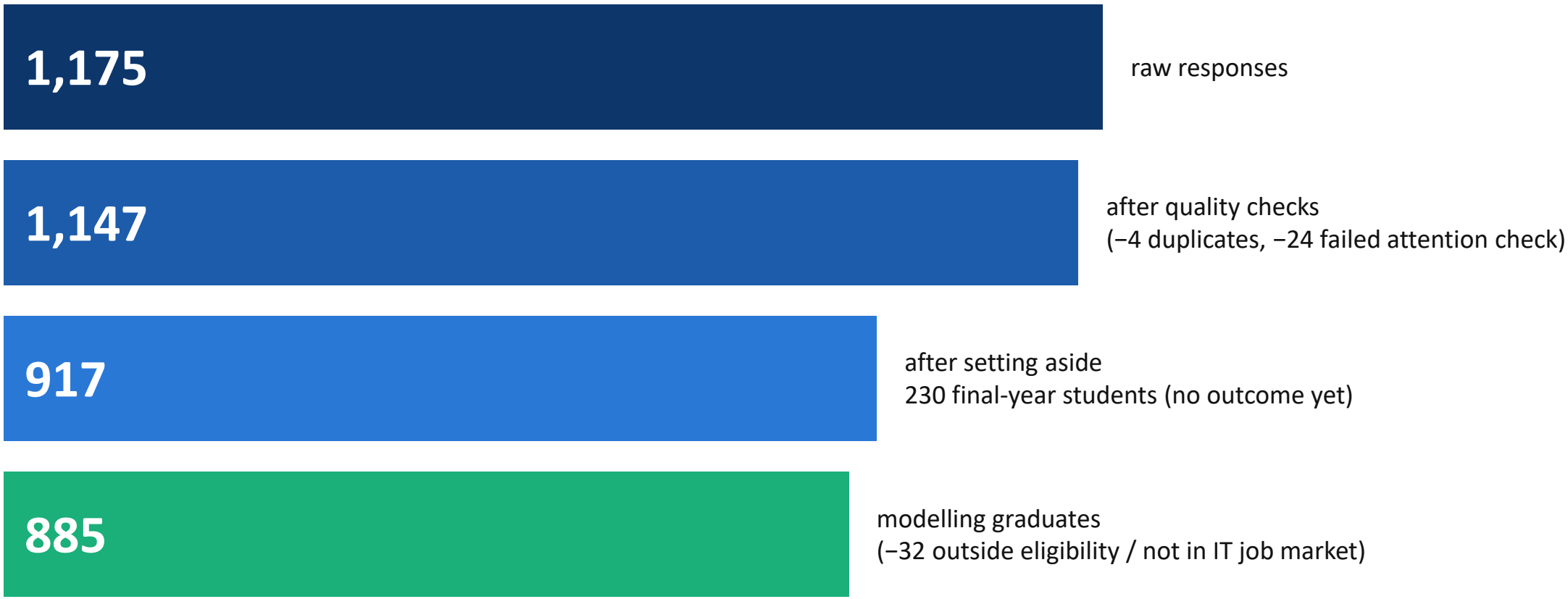
HTMT 0.68

below 0.85: technical and soft skills are distinct

- **Content validity**: 6 experts (2 lecturers, 1 survival-analysis researcher, 3 practitioners) + a per-item evidence matrix
- **Pilot**: 21 graduates and 9 employers, run before the main survey
- **Construct validity**: factor analysis (parallel analysis), AVE, Fornell-Larcker, HTMT
- **Reliability**: Cronbach's alpha with Feldt CIs, and omega; attention-check item Q18 removed 24 careless responses

From 1,175 responses to 885 modelled graduates

METHOD



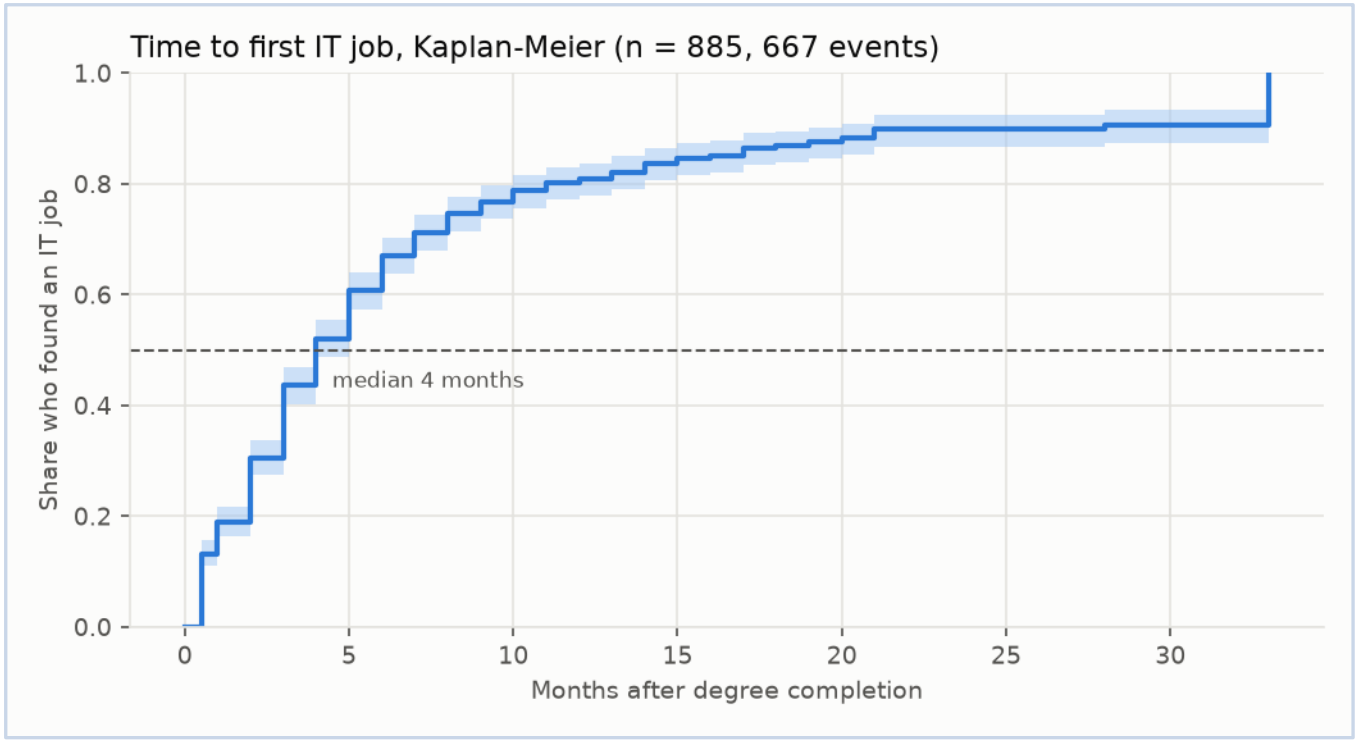
667 found a job

218 still searching

24.6% censored

What the data show: most find work within a few months

RESULTS



43.6%

found a job by 3 months

67.0%

by 6 months

80.9%

by 12 months

89.8%

by 24 months

Median 4 months · 117 graduates (13.2%) started before completion
Train / validation / test were alike: 24.7% / 24.8% / 24.2%
still searching

Seven survival models, one pre-set selection rule

MODELS

Family	Model	Why it is here
Linear	Cox Proportional Hazards	Standard baseline; hazard ratios; no distribution assumption
Regularised	Elastic Net Cox	Shrinks coefficients; guards against overfitting
Parametric	AFT: Weibull, Log-Normal, Log-Logistic	Models time directly; time ratios are easy to explain
Nonlinear	DeepSurv (neural network)	Learns nonlinear patterns; the supervisor asked for a DNN

- **Selection rule (fixed beforehand)**: highest validation Uno's C; simplicity and calibration break near-ties
- **Excluded by design**: random forests and ensembles (explainability for individuals; supervisor requirement); enforced by a test
- **Guard rails**: 29.2 events per parameter; elastic-net penalty; early stopping for DeepSurv

Evaluation protocol: the test set was used once

MODELS

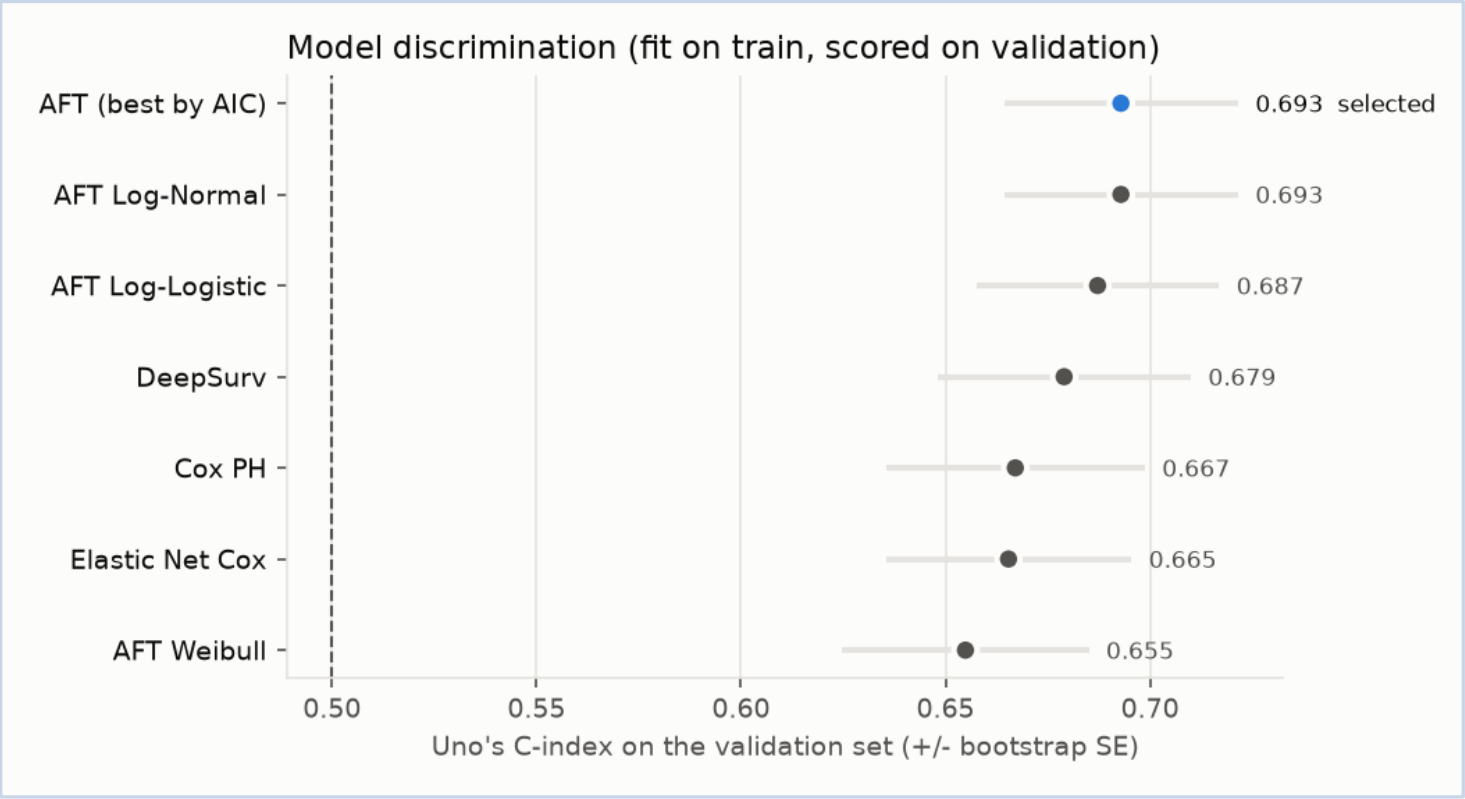
Training: 620	Validation: 133	Test: 132
fit models, tune by cross-validation	choose the model	used ONCE

- **Stratified 70 / 15 / 15**: by outcome; no graduate in two parts
- **No leakage**: imputation, scaling, encoding learned on training only
- **Test locked**: evaluated once; never used to choose or tune

Metric	What it tells you
Uno's C-index	ranking: who finds work first (0.5 = chance)
Integrated Brier score	error of predicted probabilities
Calibration (3/6/12 m)	do predicted % match reality?
Fairness, robustness	by gender, institution type, data choices

Validation: AFT Log-Normal selected (Uno's C 0.693)

RESULTS



Model	Uno's C	IBS
AFT Log-Normal ✓	0.693	0.171
AFT Log-Logistic	0.687	0.171
DeepSurv	0.679	0.175
Cox PH	0.667	0.179
Elastic Net Cox	0.665	0.175
AFT Weibull	0.655	0.179

Validation set, 133 graduates. Intervals overlap, so simplicity and calibration decide near-ties.

0.693 is VALIDATION (used to choose the model). The unbiased TEST result is 0.619, next slide.

The locked test result: honest and moderate

RESULTS

0.619

Uno's C (95% CI 0.557-0.683)

~62/100

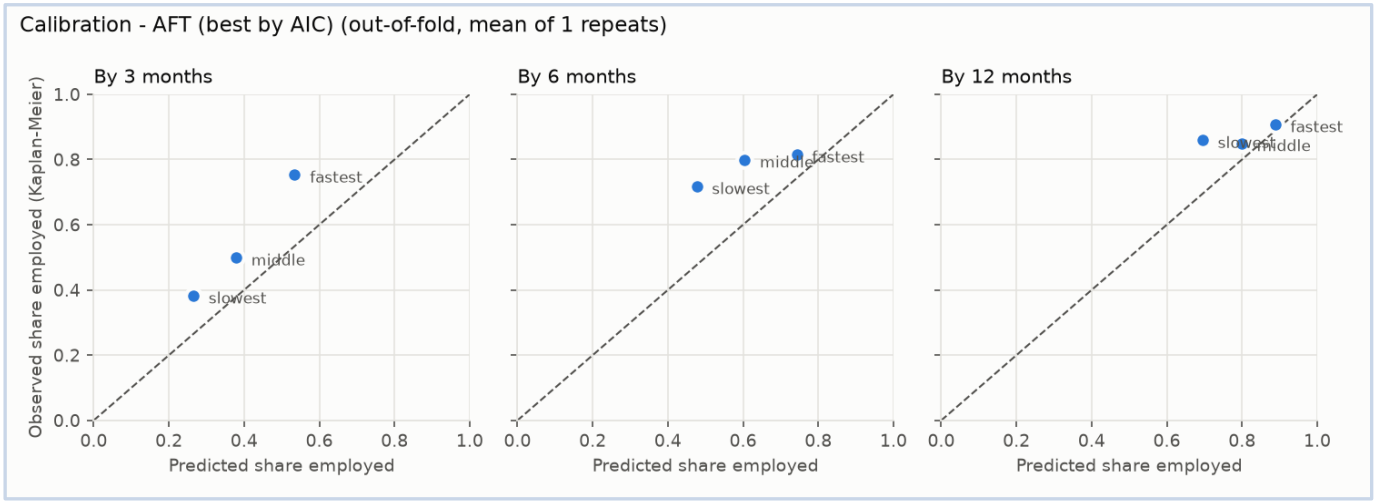
pairs ranked correctly (Harrell 0.626)

0.167

IBS (Kaplan-Meier baseline 0.164)

–17 pts

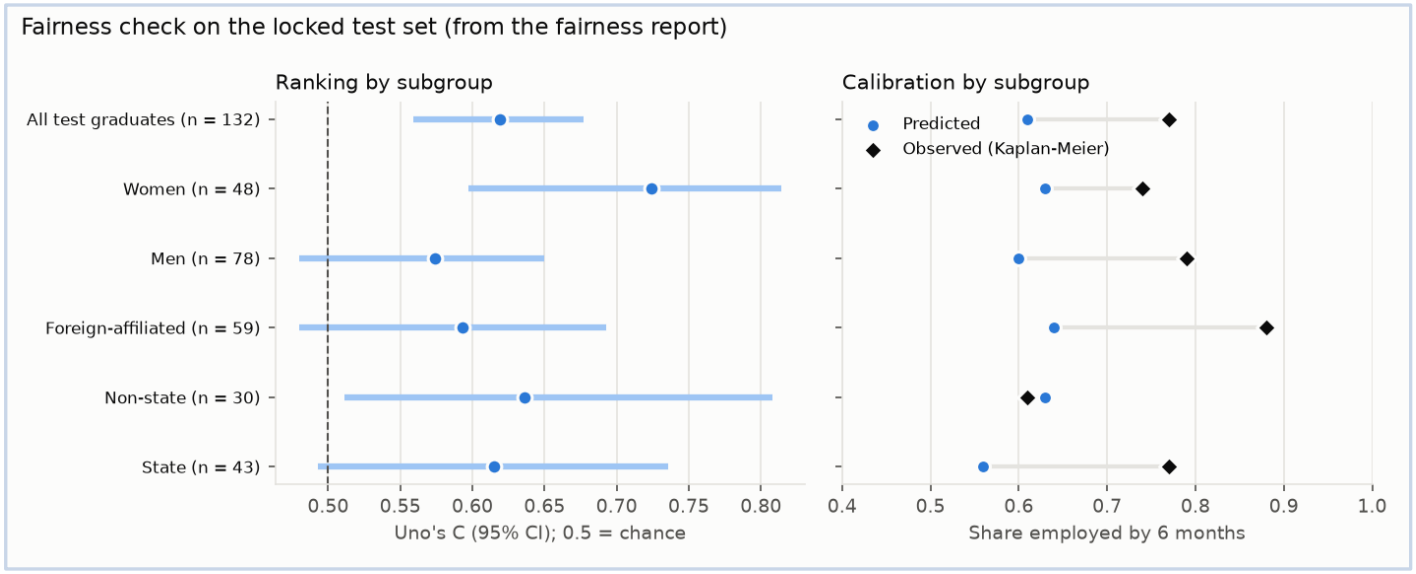
predicted 61% vs observed 77% by 6 months

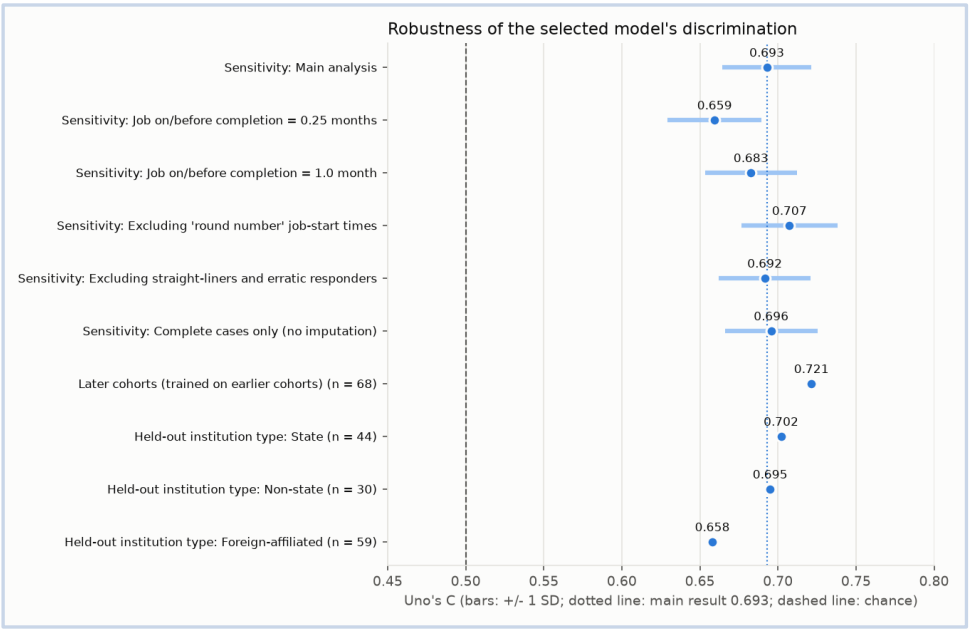


- **Why 0.619, not 0.693**: 0.693 was the validation score used to CHOOSE the model; 0.619 (about 0.62) is the unbiased test score. A drop is expected, and the intervals overlap
- **Reading it**: the ranking works; the probabilities are too pessimistic (small sets, faster later cohorts)
- **What I did**: reported it, did not re-tune on the test set; the app shows a range first and labels probabilities as indicative

Is it fair and robust? Checks on the locked test set

RESULTS





0.724 vs 0.574

women vs men Uno's C; intervals overlap; gender never a model input

0.659 - 0.707

validation C across data-cleaning variants (main 0.693)

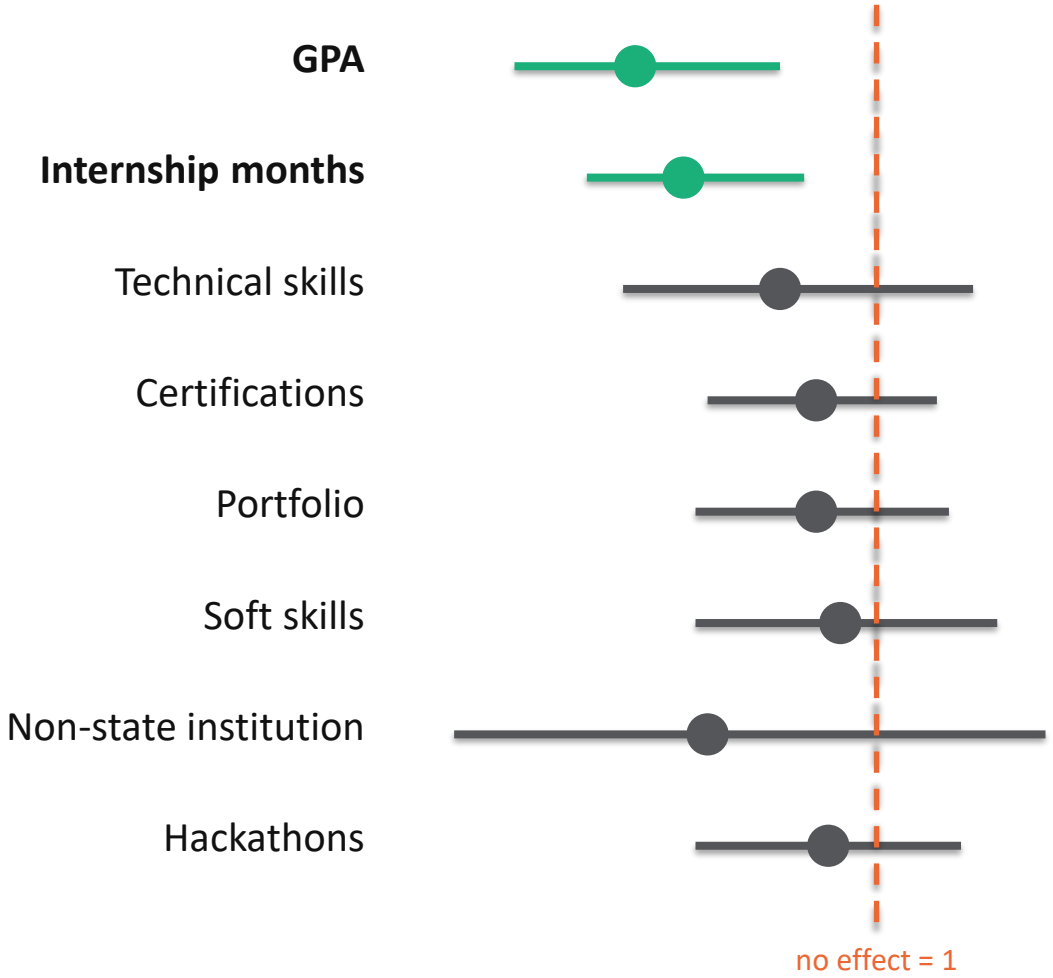
C 0.72

trained on earlier cohorts, tested on later ones

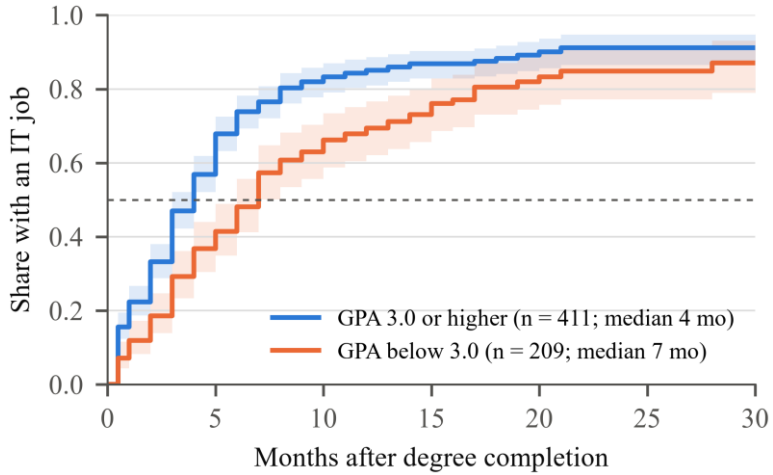
RQ1: GPA and internships are linked to a faster first job

RESULTS

Time ratio per 1 SD (95% CI): left of 1 = shorter search



GPA	**0.80**
Internship months	**0.84**
Technical skills	0.92
Certifications	0.95
Portfolio	0.95
Soft skills	0.97
Non-state institution	0.86
Hackathons	0.96



Kaplan-Meier by GPA: median 4 months (GPA ≥ 3.0) vs 7 months (below)

−20%

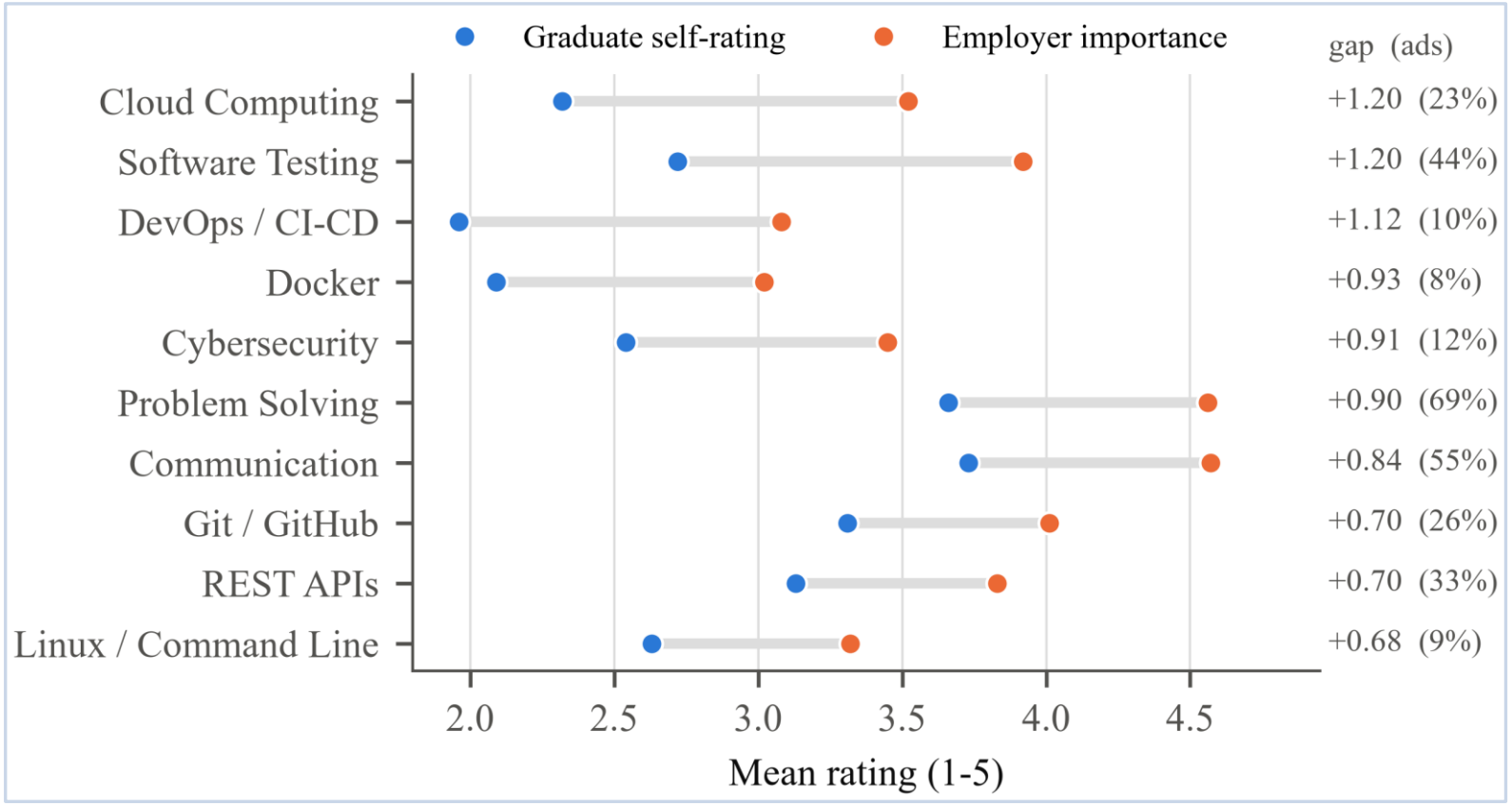
search per SD of GPA

−16%

per SD of internship

RQ2: graduates fall short of employers on 25 of 26 skills

RESULTS



25 of 26

skills where graduates rate themselves below employers

rho 0.39

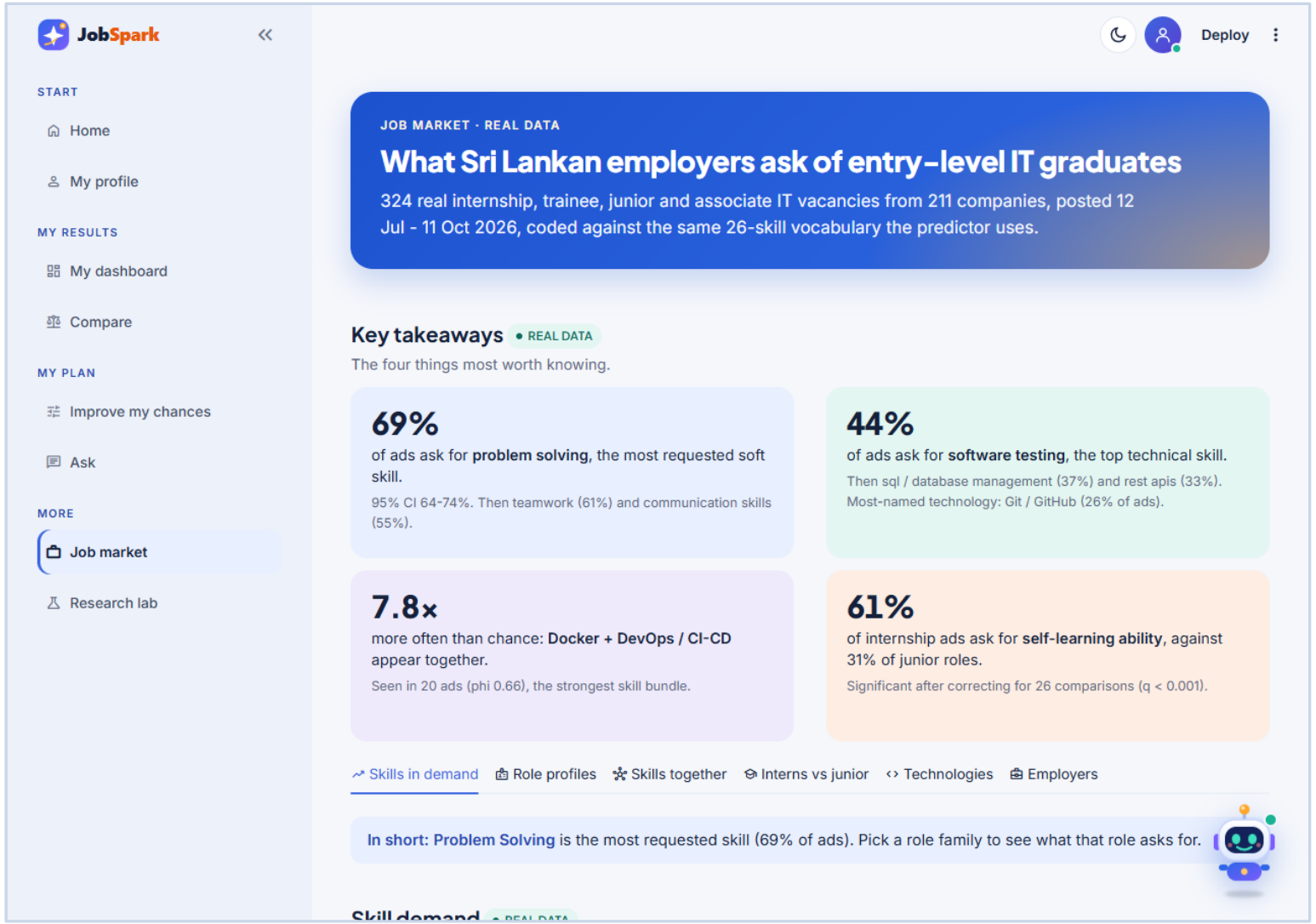
employers' stated priorities vs what ads actually ask for

3.21 / 5

employers rate GPA the LEAST important, yet it is the strongest link to speed

Real job-ad evidence: what employers actually advertise

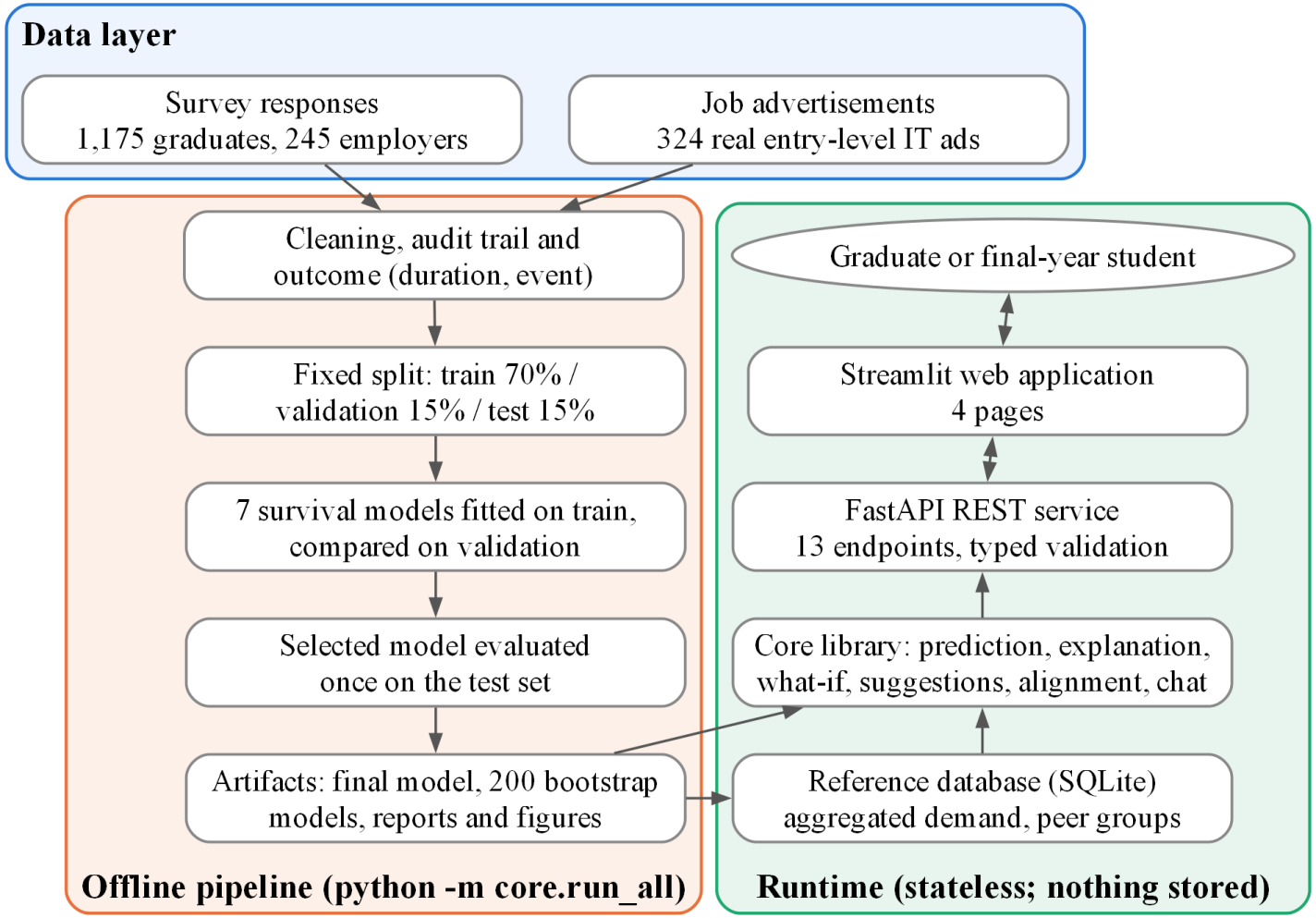
RESULTS



- **324 real ads**, 211 companies, 7 role families
- **Soft skills lead**: problem solving 69%, teamwork 61%
- **Top technical**: software testing 44%, SQL 37%
- **Skill bundle**: Docker + DevOps seen 7.8× above chance
- **Interns**: ads stress self-learning, 61% vs 31%

The artifact: an offline pipeline and a stateless app

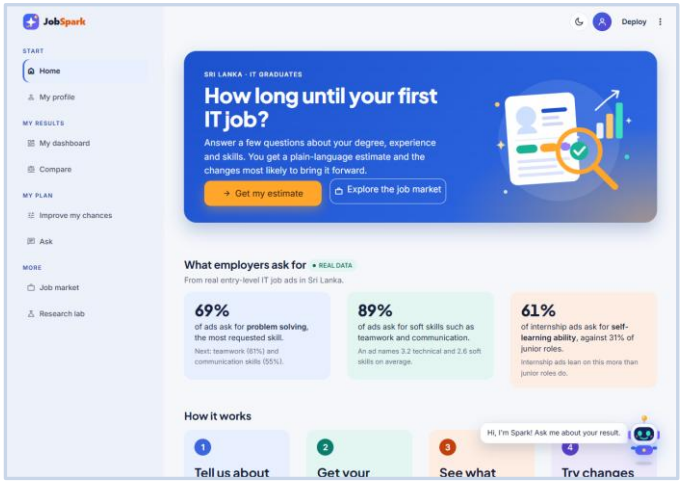
ARTIFACT



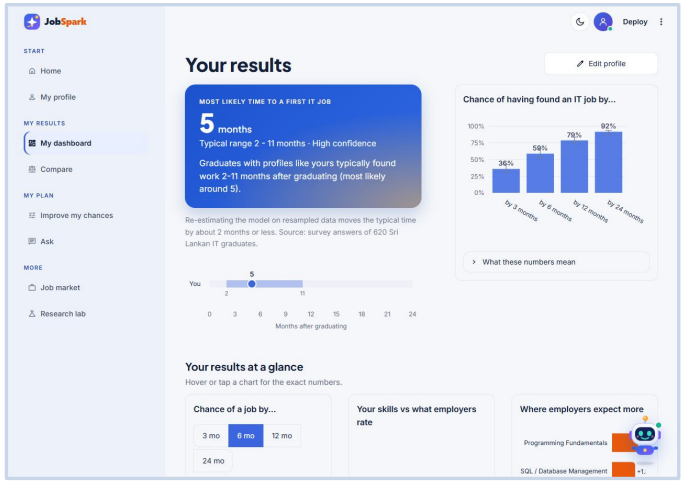
- **Offline pipeline**: clean → split → fit 7 models → select → test once → 200 bootstrap fits
- **Runtime**: FastAPI (13 endpoints) + Streamlit; loads stored results
- **Privacy by design**: no profile is stored (a test checks it)
- **Reproducible**: one command re-runs everything; data fingerprinted

The working prototype: JobSpark, live and tested

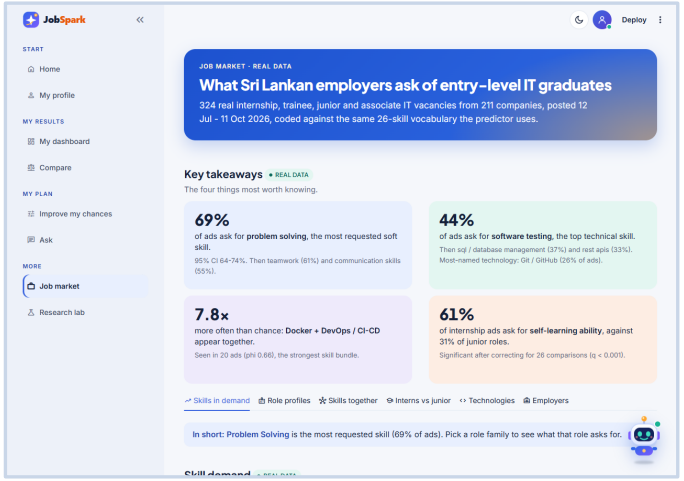
ARTIFACT



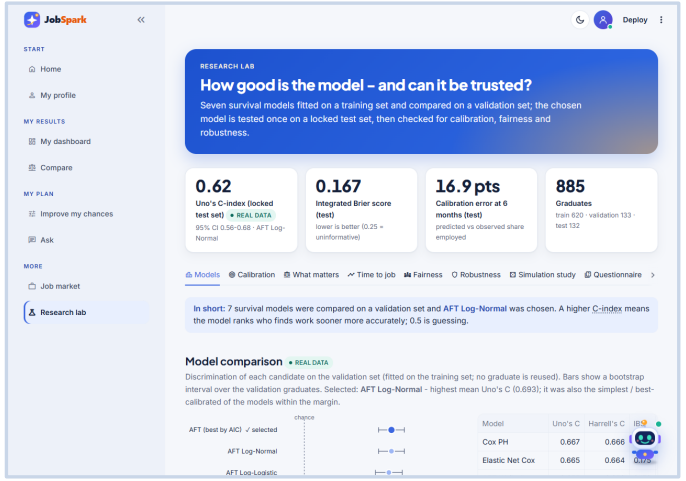
1 Home | purpose, headline job-ad figures, steps to an estimate



2 My dashboard | most likely time, range, confidence, chance of a job



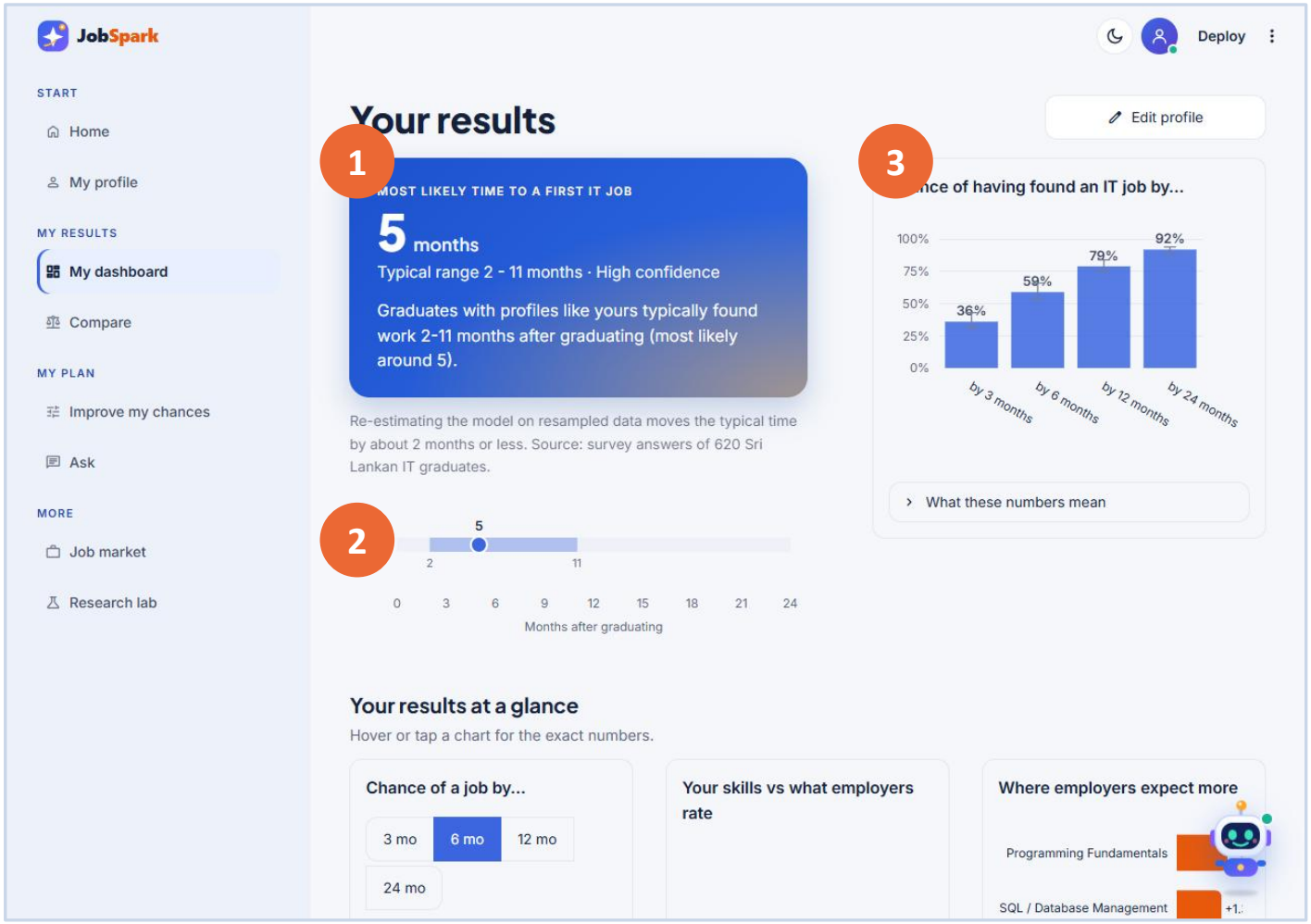
3 Job market | 324 real ads: skill demand, bundles, roles



4 Research lab | model comparison, calibration, fairness, robustness

The dashboard: a range, the chance of a job and a plan

ARTIFACT



1 **Most likely time + range + confidence**

'5 months, typical range 2-11, high confidence'

2 **Range bar**

the middle half of similar graduates, at a glance

3 **Chance of a job by 3 / 6 / 12 / 24 months**

bars with intervals; hover for the exact numbers

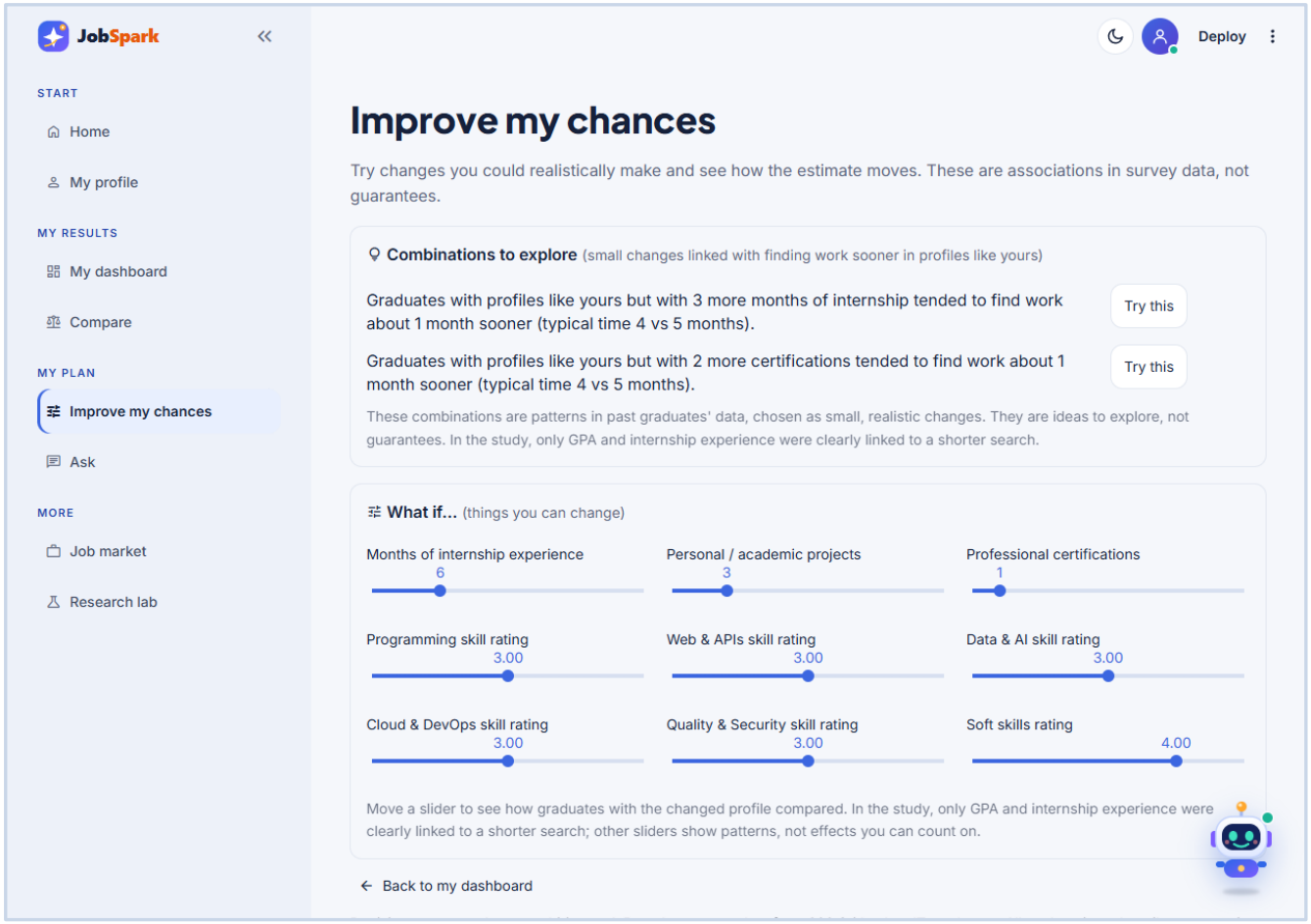
4 **Results at a glance**

chance of a job, skills against what employers rate, where employers expect more

Redesigned after the usability study: step-by-step profile, dashboard, light and dark themes. The model, the API and the results are unchanged.

Combinations to explore, and what-if sliders

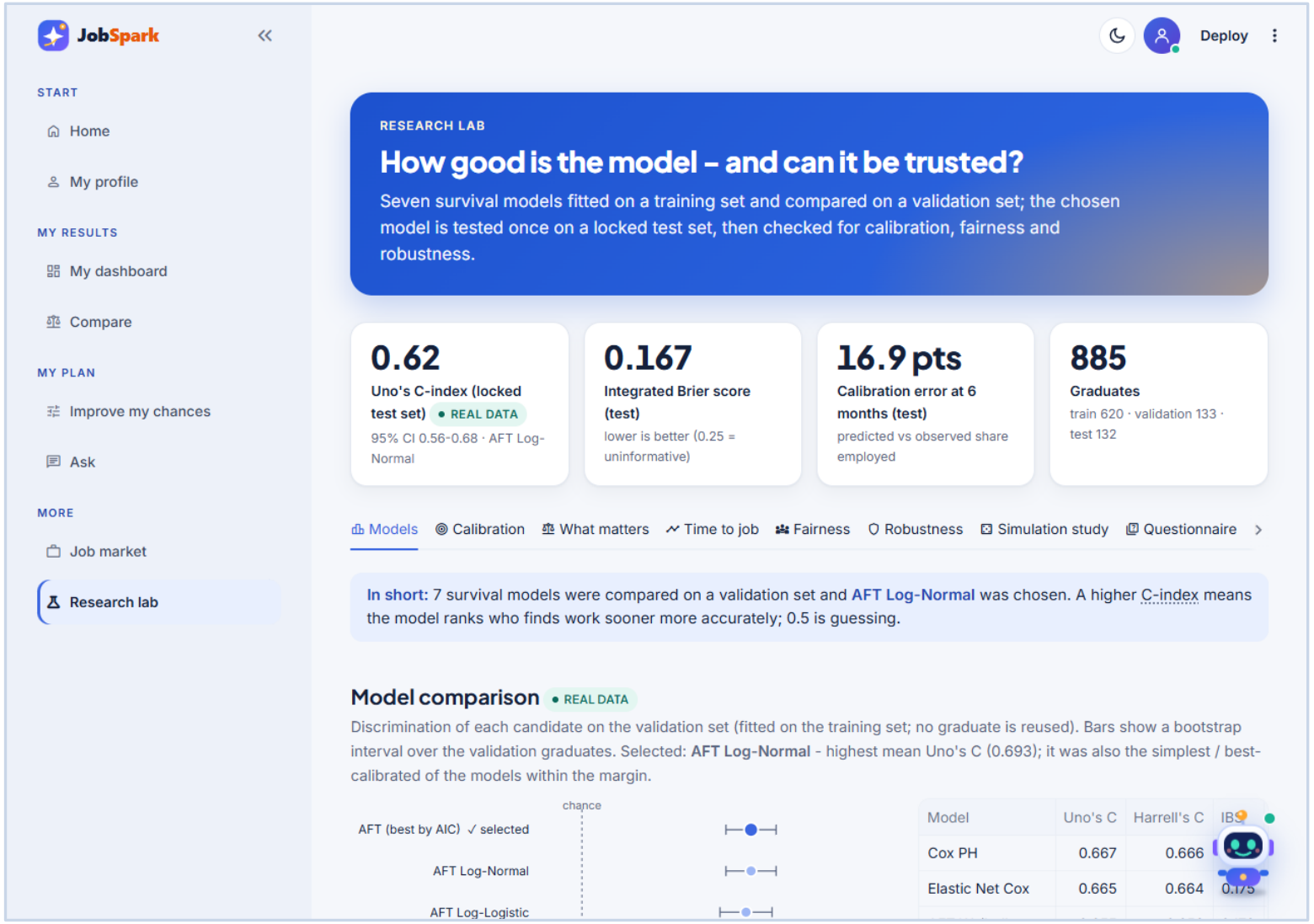
ARTIFACT



- **Combinations to explore**: small, realistic changes linked with faster searches in profiles like yours
- **What-if sliders**: change internship, projects, certifications, skill ratings and see the range move
- **Careful wording**: 'tended to', never 'will'; patterns, not causes
- **Honest**: the app says only GPA and internship were clearly linked
- **Grounded assistant (Spark)**: answers only from the system's own results; nothing stored

Transparency built in: the Research lab

ARTIFACT



- **Models**: all 7 compared, selected one marked
- **Tabs**: calibration, what matters, fairness, robustness, simulation, questionnaire
- **Real vs synthetic**: every figure is tagged
- **Honest**: shows the 16.9-point calibration error

Why not just use ChatGPT? I tested it

ARTIFACT

0.619
this model

0.539
ChatGPT (near chance)

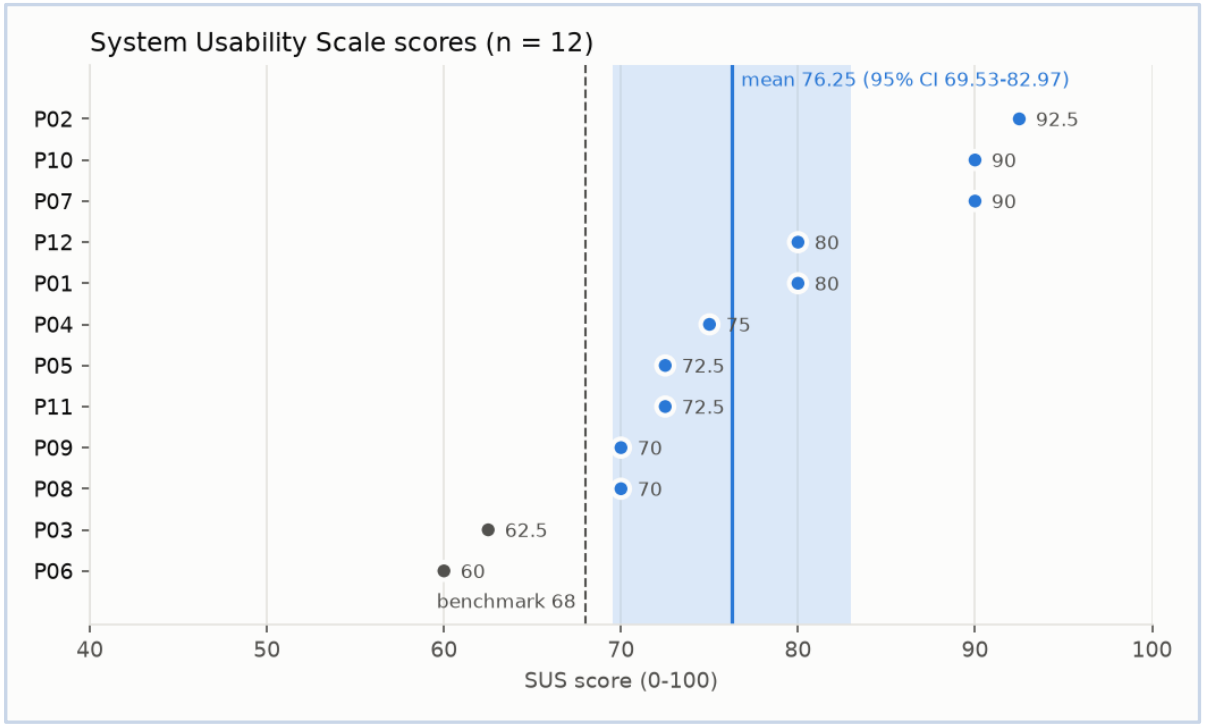
+0.081
difference, 95% CI 0.030-0.131

132
same test graduates

	This system	ChatGPT (general LLM)
Trained on	620 real Sri Lankan graduates, for this one outcome	web text, for general language
Uses graduates still searching	Yes (censoring)	No
Consistency	same profile → same answer	different answers on repeated runs
Bias on the 132 graduates	under-predicts (pessimistic)	over-optimistic within 6 months
Best at	individual, consistent, evidence-based timeline	general advice and wording

Usability: SUS 76.3 with 12 users, and what changed after

ARTIFACT



What users said	What changed
Main result not obvious (3)	Chat folded away so the estimate stays the focus
Terms and charts need explaining (3)	'How to read this result' guide added
Navigation inconsistent (2)	Page headings now match the menu
Unsure the numbers apply to them (2)	Data-source line under the estimate

76.3
mean SUS

10 / 12
scored ≥ 68

α 0.90
reliability

Honest limits: 12 peers, task completion not recorded, fixes not re-tested.

Quality and trust: tests, reproducibility, ethics

ARTIFACT

295
automated tests pass

1 command
re-runs the whole pipeline

Public
code and walkthrough notebook on GitHub

0 stored
profiles kept by the app (tested)

- **Testing**: cleaning rules, model interface, API schemas, SUS scoring, privacy guarantee, wording rules
- **Reproducibility**: fixed seeds, data fingerprint, walkthrough notebook that re-computes every reported metric
- **Ethics**: approved before collection; consent; no names or contacts; PDPA 2022; fairness checked; associational wording
- **Process**: pre-set analysis plan; every deviation documented with its reason

Limitations, stated openly

REFLECTION

Limitation	How it is handled
Non-probability sample; a third from one institution	stated; compared with Kelaniya 2022; results describe this sample
Self-reported outcome and recalled dates	month-and-year questions, impossible dates removed, sensitivity checks
One reference survey date for every response	documented deviation; may overstate follow-up for those still searching
Small evaluation sets (about 130); probabilities pessimistic	reported openly; app leads with a range; labelled indicative
Observational data	associational wording everywhere; no causal claims
Employer ratings cluster at 4-5	used to rank skills; checked against real job ads
SUS: 12 peers, no task-by-task completion	stated; fixes made but not re-tested

Answers to the research questions, and the contribution

REFLECTION

RQ1 GPA and internship months are linked to 20% and 16% shorter searches; institution and degree type show no clear link

RQ2 graduates rate themselves below employers on 25 of 26 skills; biggest gaps in cloud, testing and DevOps

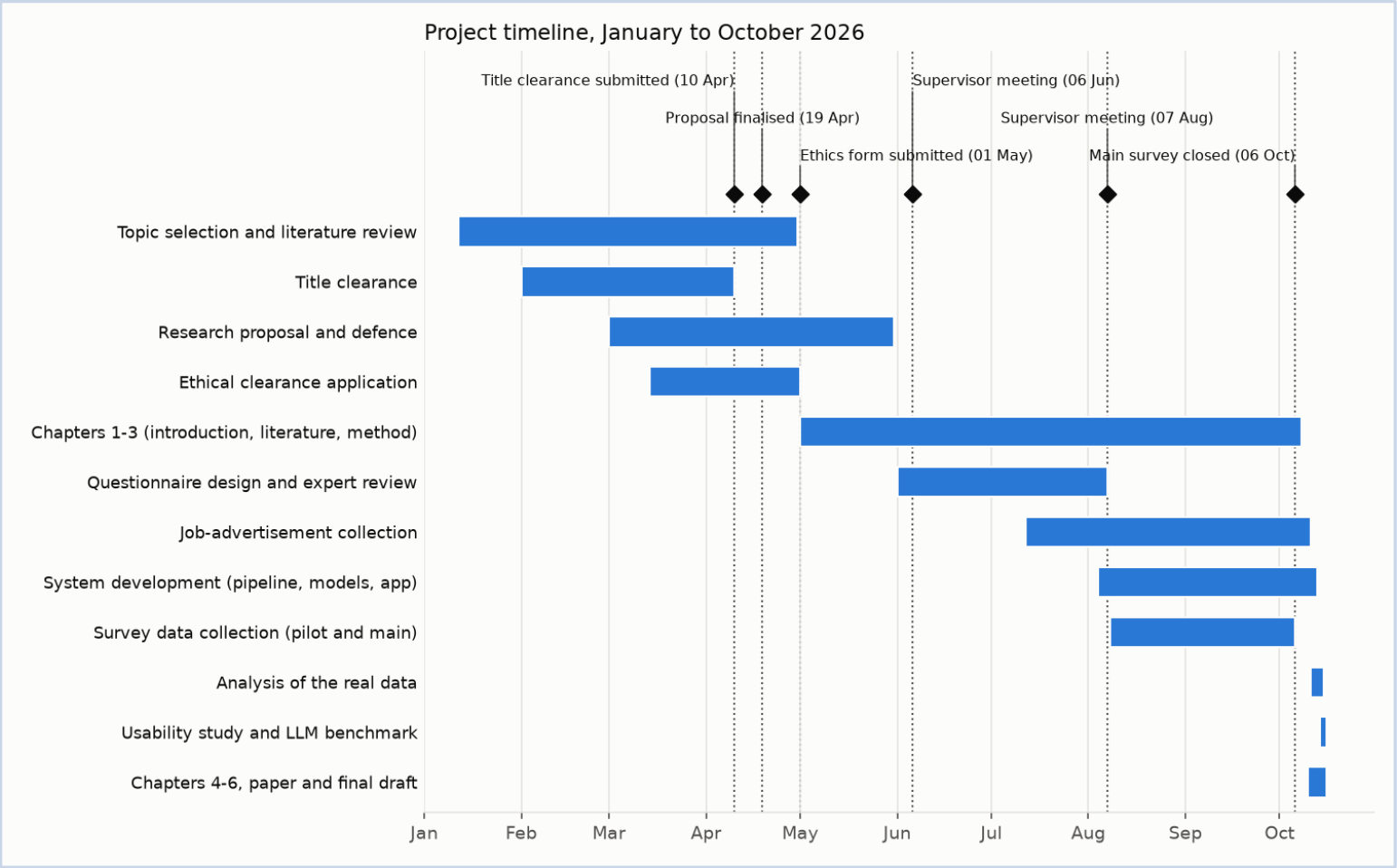
RQ3 AFT Log-Normal chosen by a pre-set rule; moderate ranking on the locked test (C 0.619); probabilities too pessimistic

RQ4 usable (SUS 76.3) and better than ChatGPT on the same graduates (0.619 vs 0.539)

Contribution: the first explainable survival-analysis system that predicts WHEN Sri Lankan IT graduates find a first IT job, linked to real employer demand and tested against a general LLM.

Project journey and what comes next

REFLECTION



Next steps

- **More data**: more respondents and institutions
- **Recalibrate**: fix the pessimistic probabilities
- **Real dates**: per-respondent survey dates
- **Re-test**: the improved app with task-level measures
- **Track outcomes**: follow graduates after using the tool

Four things to remember

WHEN, not just whether

survival analysis uses the one in four still searching

Honest evidence

locked test set used once: C 0.619, reported with its flaws

Beats ChatGPT

0.619 vs 0.539 on the same 132 graduates

Actually built

working app, SUS 76.3, 295 tests, public code

Thank you

Questions and discussion

M.S.F. Shazna · Supervisor: L. S. Chathurika
Code, tests and walkthrough notebook: github.com/ShaznaSalman/final-research-employability

Backup: the hard questions, ready

BACKUP

'0.62 is weak'

0.62 is the unbiased TEST score (0.693 was the validation score used to choose the model). Honest, not tuned; contacts and interviews weren't measured; more data adds ~0.01-0.02.

'It under-predicts'

Yes, by up to 17 pts. Small sets and faster later cohorts. Reported, not tuned away; the app leads with a range.

'Why not ChatGPT?'

Tested on the same 132 graduates: 0.539 vs 0.619, and inconsistent across runs.

'Will it get me a job?'

No promise. It shows a realistic timeline and which changeable factors went with faster hiring.

'Biased sample?'

Non-probability, stated. Compared with Kelaniya 2022; results describe this sample.

'Why no Random Forest?'

Individual explanations need interpretable time ratios; supervisor requirement; a test enforces it.

Backup: key numbers

BACKUP

Measure	Value	Measure	Value
Graduates / employers / ads	1,175 / 245 / 324	Split	620 / 133 / 132
Modelled graduates	885 (667 found a job, 218 searching)	Selected model	AFT Log-Normal
Validation Uno's C (chose the model)	0.693 (0.636-0.745)	TEST Uno's C (headline, say 0.62)	0.619 (0.557-0.683)
Test Harrell's C / IBS	0.626 / 0.167 (KM 0.164)	ChatGPT Uno's C	0.539 (diff 0.081)
GPA / internship time ratio	0.80 / 0.84 per SD	Events per parameter	29.2
Found a job by 3/6/12/24 m	43.6 / 67.0 / 80.9 / 89.8%	Median time	4 months
SUS	76.3 (69.5-83.0), 10/12 ≥ 68	Cochran target	n = 384